will one day halt, reverse, and the universe will implode, ending in the so-called big crunch. Calculations show that even though the size of the universe would decrease during the implosion phase, entropy would continue to rise, which demonstrates that small size does not ensure low entropy. In Chapter 11, though, we will see that the universe's small initial size does play a role in our current, best explanation of the low entropy beginning.

Chapter 7

1. It is well known that the equations of classical physics cannot be solved exactly if you are studying the motion of three or more mutually interacting bodies. So, even in classical physics, any actual prediction about the motion of a large set of particles will necessarily be approximate. The point, though, is that there is no fundamental limit to how good this approximation can be. If the world were governed by classical physics, then with ever more powerful computers, and ever more precise initial data about positions and velocities, we would get ever closer to the exact answer.

2. At the end of Chapter 4, I noted that the results of Bell, Aspect, and others do not rule out the possibility that particles always have definite positions and velocities, even if we can't ever determine such features simultaneously. Moreover, Bohm's version of quantum mechanics explicitly realizes this possibility. Thus, although the widely held view that an electron doesn't have a position until measured is a standard feature of the conventional approach to quantum mechanics, it is, strictly speaking, too strong as a blanket statement. Bear in mind, though, that in Bohm's approach, as we will discuss later in this chapter, particles are "accompanied" by probability waves; that is, Bohm's theory always invokes particles *and* waves, whereas the standard approach envisions a complementarity that can roughly be summarized as particles *or* waves. Thus, the conclusion we're after— that the quantum mechanical description of the past would be thoroughly incomplete if we spoke exclusively about a particle's having passed through a unique point in space at each definite moment in time (what we *would* do in classical physics)—is true nevertheless. In the conventional approach to quantum mechanics, we must also include the wealth of other locations that a particle could have occupied at any given moment, while in Bohm's